EXPERIMENT 8

Question:

Write the Python program to implement DFS.

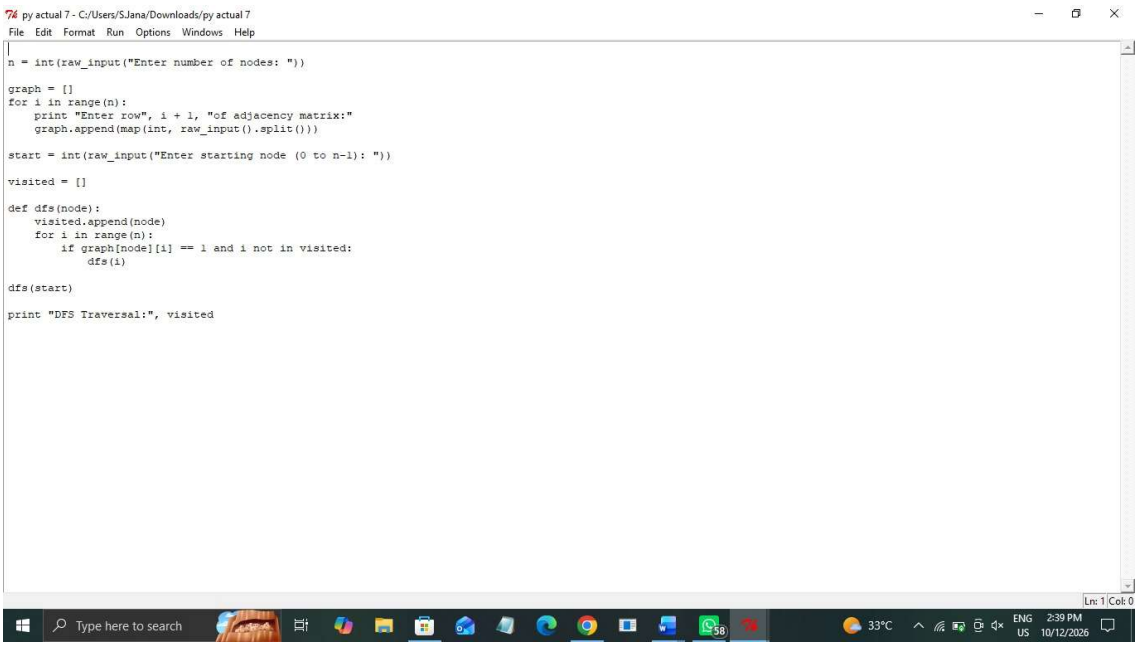
Aim

To write a Python program to implement the Depth First Search (DFS) algorithm to traverse a graph.

Algorithm

1. Start the program.
2. Create a graph using a dictionary.
3. Start DFS from the given node.
4. Mark the node as visited and print it.
5. Visit each unvisited neighboring node recursively.
6. Repeat until all reachable nodes are visited.
7. Stop the program.

Code:



Output:

```
>>>
Enter number of nodes: 4
Enter row 1 of adjacency matrix:
0 1 1 0
Enter row 2 of adjacency matrix:
1 0 0 1
Enter row 3 of adjacency matrix:
1 0 0 1
Enter row 4 of adjacency matrix:
0 1 1 0
Enter starting node (0 to n-1): 0
DFS Traversal: [0, 1, 3, 2]
>>>
```

Result

Thus, the Python program to implement the Depth First Search (DFS) algorithm was executed successfully, and all reachable nodes of the graph were visited.